

TrES-5b

R-1778

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_220715 · created 2026-08-02T22:57:43+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459775.8657 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.045 +/- 0.053
Transit depth (Rp/Rs)^2	0.21 +/- 0.48 [%]
Orbital Inclination (inc)	80.33 +/- 2.91
Airmass coefficient 1 (a1)	30.55 +/- 16.65
Airmass coefficient 2 (a2)	-3.0 +/- 1.36
Scatter in the residuals of the lightcurve fit is	51.4 %
Best Comparison Star	#3 - [452, 128]
Optimal Aperture	2.82
Optimal Annulus	7.7
Transit Duration (day)	0.026 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.045 ± 0.053 vs 0.142 (deviation 1.72σ)
Duration fit vs published	0.026 d vs 0.0758 d (deviation 1.74σ)
Mid-time O–C	8.1 min
Depth SNR	0.8
Residual scatter	51.4 %

Light curve

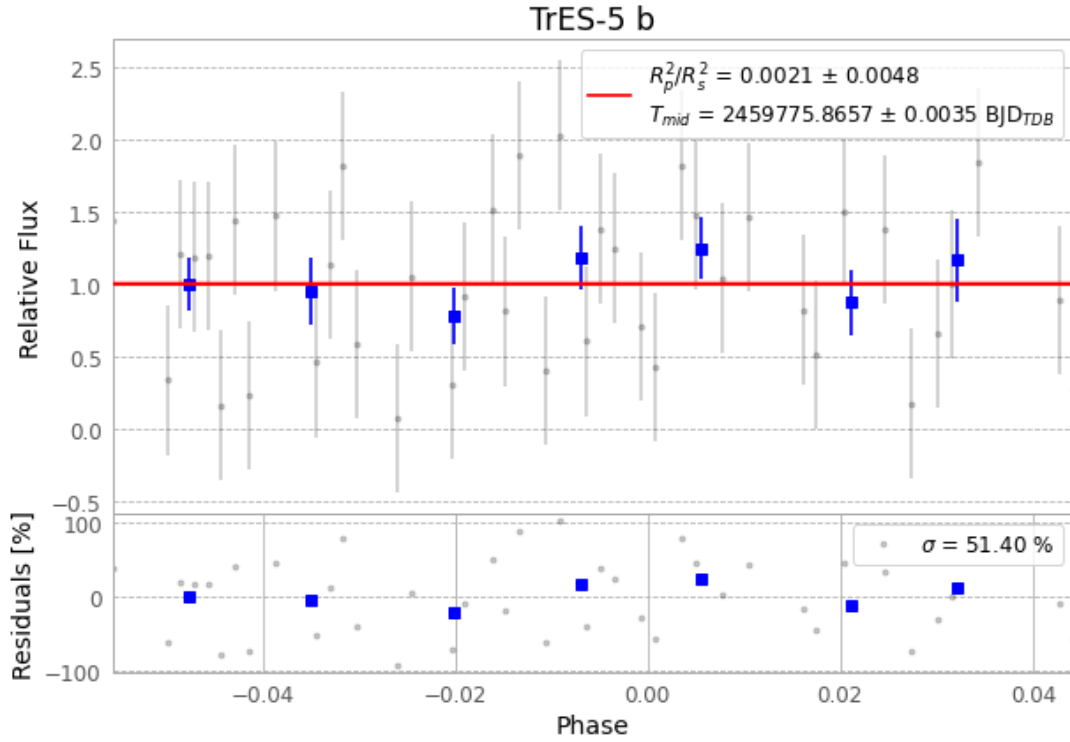


Figure 1 — FinalLightCurve_TrES-5 b_2022-07-14.png (SHA-256 c056df75668c164b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [493, 168], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4be3f80ada1fb835...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), TRES-5220715064513.FITS → TRES-5220715103313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-220715.log (cb54c22b6671...), FinalLightCurve_TrES-5 b_2022-07-14.png (c056df75668c...), FinalLightCurve_TrES-5 b_2022-07-14.csv (ca354658408b...)

Compute resources

Wall time 195.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 22:57Z.